COMPUTER SCIENCE TRIPOS Part IB, Part II 50% – 2022 – Paper 7

4 Formal Models of Language (pjb48)

This question concerns a hypothesis space where each language is characterised by three binary parameters:

Parameter name	0	1
S_{right}	subject before verb	subject after verb
O_{right}	objects before verb	objects after verb
S_{drop}	subject is required	subject is optional

models of languages & language patterns

(*a*) Complete the table of language patterns exhibited by all possible parameter combinations. Denote S-optionality with (S). When the order of S and O is ambiguous, place O closest to the verb. The first two rows are given:

S_{right}	O_{right}	S_{drop}	language pattern
0	0	0	SOV
0	0	1	(S)OV

[3 marks]

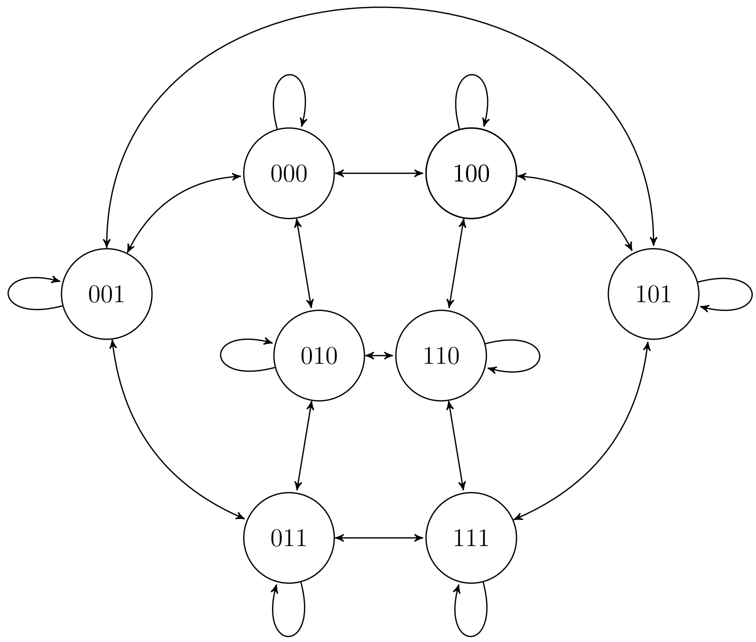
Answer: 0.5 for each remaining row in the table.

S_{right}	O_{right}	S_{drop}	language pattern
0	0	0	SOV
0	0	1	(S)OV
0	1	0	SVO
0	1	1	(S)VO
1	0	0	OVS
1	0	1	OV(S)
1	1	0	VOS
1	1	1	VO(S)

a learning algorithm within a learning paradigm

(*b*) A learning algorithm attempts to identify the parameter settings which characterise a *target language*. The algorithm records the *learner state* (the current hypothesised parameter settings). It proceeds by checking the compatibility of an input sentence with the learner state. If they are not compatible, the algorithm flips **one** bit in the learner state. Draw a diagram to represent the possible transitions between learner states. [2 marks]

Answer: 1 mark for showing that there are 3 possible transitions from each node in order to randomly flip one bit. 1 mark for the self-loop indicating a compatible input sentence.



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a learning algorithm

(*c*) The target language is represented by the parameters [011] ($S_{\text{right}} = 0$, $O_{\text{right}} = 1$, $S_{\text{drop}} = 1$). Sentences in a language may contain 0, 1 or 2 objects and therefore sentences in the target language are in one of the following forms:

1. SV 2. SVO 3. SVOO 4. V 5. VO 6. VOO

When presented to the learning algorithm, the probability distribution over sentence forms is uniform. Each bit has an equal probability of being selected for flipping when the input is not compatible. Provide a transition matrix showing transition probabilities between learner states. [4 marks]

Answer: Table below shows when input will be compatible for each learner state (probability provides diagonal for required transition matrix). This table is not required. Produced here for clarity.

Parameters	Pattern	Possible forms	Compatible with input
000	SOV	**SV** SOV SOOV	1/6
001	(S)OV	**SV** SOV SOOV **V** OV OOV	2/6
010	SVO	**SV SVO SVOO**	3/6
011	(S)VO	**SV SVO SVOO V VO VOO**	6/6
100	OVS	VS OVS OOVS	0/6
101	OV(S)	VS OVS OOVS **V** OV OOVS	1/6
110	VOS	VS VOS VOOS	0/6
111	VO(S)	VS VOS VOOS **V VO VOO**	3/6

Transition matrix below. Fiddly so mistakes tolerated. 1 mark for evidence of considering the input forms. 1 mark for dividing the remainder of probability by 3 possible transitions. 1 mark for non-symmetrical table. 1 mark for the from 011 line.

From↓, To→	000	001	010	011	100	101	110	111
000	$\frac{1}{6}$	$\frac{5}{6}\times\frac{1}{3}$	$\frac{5}{6}\times\frac{1}{3}$	0	$\frac{5}{6}\times\frac{1}{3}$	0	0	0
001	$\frac{4}{6}\times\frac{1}{3}$	$\frac{2}{6}$	0	$\frac{4}{6}\times\frac{1}{3}$	0	$\frac{4}{6}\times\frac{1}{3}$	0	0
010	$\frac{3}{6}\times\frac{1}{3}$	0	$\frac{3}{6}$	$\frac{3}{6}\times\frac{1}{3}$	0	0	$\frac{3}{6}\times\frac{1}{3}$	0
011	0	0	0	1	0	0	0	0
100	$\frac{1}{3}$	0	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0
101	0	$\frac{5}{6}\times\frac{1}{3}$	0	0	$\frac{5}{6}\times\frac{1}{3}$	$\frac{1}{6}$	0	$\frac{5}{6}\times\frac{1}{3}$
110	0	0	$\frac{1}{3}$	0	$\frac{1}{3}$	0	0	$\frac{1}{3}$
111	0	0	0	$\frac{3}{6}\times\frac{1}{3}$	0	$\frac{3}{6}\times\frac{1}{3}$	$\frac{3}{6}\times\frac{1}{3}$	$\frac{3}{6}$

learning algorithm evaluation

(*d*) If the algorithm is initialised with state [100], give an equation for the expected number of steps to converge on the target language parameters. [3 marks]

Answer:

$$N_{100} = 1 + T(100,100)N_{100} + T(100,000)N_{000} + T(100,101)N_{101} + T(100,110)N_{110}$$

where N_x is the expected number of steps from state x
and $T(x,y)$ is the transition probability from x to y.
1 mark for N, 1 for T, 1 for the equation.

learning algorithm evaluation

(*e*) Given equal probability of any given state being used to initialise the algorithm, write an equation for the overall expected number of steps before convergence on the target language parameters. [2 marks]

Answer: Given transition matrix above, some starting states will have quicker expected convergence than others. We have equal chance of starting in any state so consider expected steps from each starting state and divide by total number of states. $N = \frac{\sum_{x \in allstates} N_x}{8}$
1 mark for summing over states. 1 mark for dividing by number of states.

criteria for success & human language learning

(*f*) Suggest an alternative learning algorithm within the same learning paradigm and discuss the assumptions of the learning paradigm with respect to human language learning. [6 marks]

Answer: 2 marks for a suggested alternation to the algorithm [1 mark] that doesn't alter the paradigm [1 mark]. 4 remaining marks for discussion of the paradigm. 1 mark for each relevant point. e.g. mention of positive evidence only (no errors or negative evidence), unlimited time for convergence, attempting to learn from each utterance, limitations of the hypothesis space, the implicit assumptions of the criteria for success, risk of non-convergence for certain target languages.